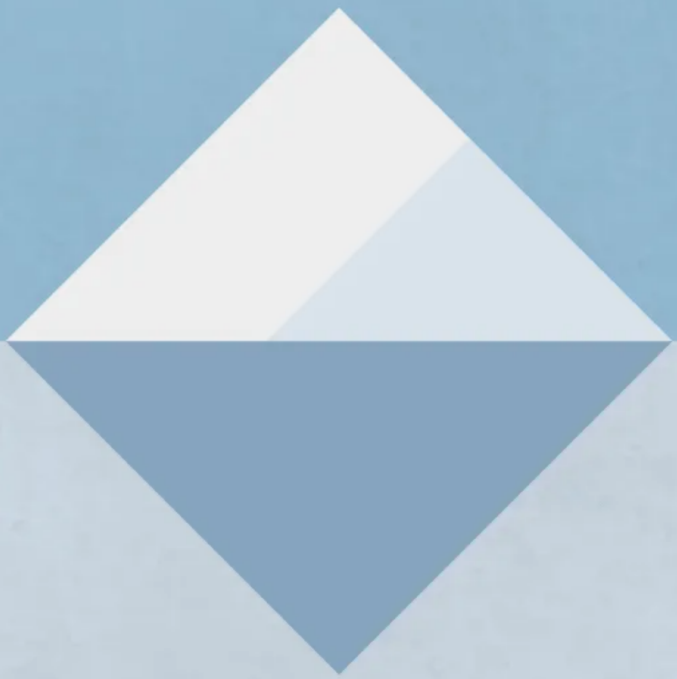
NOTEBOOK



1. 量词转换原则

$$\neg \forall x\, P(x) \equiv \exists x\, \neg P(x)$$

$$\neg \exists x\, P(x) \equiv \forall x\, \neg P(x)$$

$$\neg (A \wedge B) \equiv \neg A \vee \neg B$$

$$\neg (A \vee B) \equiv \neg A \wedge \neg B$$

$$\neg (A \rightarrow B) \equiv A \wedge \neg B$$

증명 $\forall x(\exists y P(x,y) \to \forall z Q(x,z))$

$\to \neg[\forall x(\exists y P(x,y) \to \forall z Q(x,z))]$

$\to \exists x \neg(\exists y P(x,y) \to \forall z Q(x,z))$

$\to \exists x(\exists y P(x,y) \wedge \neg \forall z Q(x,z))$

$\to \exists x(\exists y P(x,y) \wedge \exists z \neg Q(x,z))$

二部图：

$$\begin{tikzcd} & U \\ V \arrow[ur] \arrow[dr] & \\ & V' \end{tikzcd}$$

所有的边都是 $U \to V'$ 或 $V' \to U$.

奇偶的冲突：

奇数：出发点的对面

偶数：回到原来的集

环

奇数环不成立.

$$A \vee (A \wedge B) = A$$

$$A \wedge (A \vee B) = A$$

$$A \vee (\neg A \wedge B) = A \vee B$$

$$A \wedge (\neg A \vee B) = A \wedge B$$

$$(P \vee Q) \wedge ((P \vee Q) \vee R) = P \vee Q$$

$$\sum_{v \in V} \deg(v) = 2|E|$$

$|E| = 15$

设顶点数为 n，每个顶点的度数为 d

和：$n \cdot d$

$n \cdot d = 30$

猜测：

1	30	×
2	14	×
3	10	×
4	6	×
5	5	✓
10	3	✓
11	2	✓
12	1	✓

完全数